

Alexander Ho

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Personal Information

Citizenship 📌 United States

Current Work Address 📌 University of Illinois Urbana-Champaign Beckman Institute for Advanced Science and Technology 405 N Mathews Ave. Urbana, IL 61801

Research Interests 📌 Optical Coherence Tomography, Multiphoton Microscopy, Raman Microscopy, Fluorescence Microscopy, Image Processing, Computational Modeling, Machine Learning, Deep Learning, Medical Device Development, Early Disease Detection, Translational Medicine

Education

2021 – 📌 **Ph.D. in Bioengineering** University of Illinois Urbana-Champaign

2021 – 2025 📌 **M.S. in Bioengineering** University of Illinois Urbana-Champaign

2014 – 2018 📌 **B.S. in Biomedical Engineering** Tufts University

Employment History

2021 – 📌 **Graduate Research Assistant, Boppart Lab**
Beckman Institute for Advanced Science and Technology,
University of Illinois Urbana-Champaign (Urbana, IL, United States)

2018 – 2021 📌 **Research Technician II, Evans Lab**
Wellman Center for Photomedicine, Massachusetts General Hospital,
Harvard Medical School (Boston, MA, United States)

Research Publications

Manuscripts in Prep

- 1 K. Bae, M. Özbey, **A. Ho**, *et al.*, *Label-free segmentation of mitochondria for simultaneous morphological and metabolic studies*, submitted.
- 2 H. Fan, J. Shi, Z. Yang, *et al.*, *Cophas: Structure-anchored confidence-domain for label-free metabolic imaging*, in prep.
- 3 S. Fitzgerald, **A. Ho**, G. L. Monroy, J. A. Werkhaven, S. A. Boppart, and A. Mahadevan-Jansen, *Noninvasive characterization of otitis media with image-guided raman spectroscopy*, in prep.
- 4 **A. Ho**, S. Fitzgerald, G. L. Monroy, *et al.*, *A handheld probe combining optical coherence tomography(oct) and raman spectroscopy (rs): A feasibility study in vivo*, in prep.
- 5 **A. Ho**, J. Shi, E. Chaney, *et al.*, *Predicting productivity in cho cells based on co-occurrence of metabolism-related features in simultaneous label-free autofluorescence multiharmonic (slam) microscopy*, in prep.
- 6 J. Shi, **A. Ho**, J. E. Sorrells, *et al.*, *Machine learning-assisted early biopharmaceutical cho cell line selection using label-free multimodal nonlinear optical microscopy*, submitted.

Peer-Reviewed Journal Publications

- 1 J. E. Kerschner, R. A. Daley, R. Link, *et al.*, “Clinical trial assessing the use of optical coherence tomography in decision making for children presenting with acute otitis media,” *International Journal of Pediatric Otorhinolaryngology*, vol. 202, p. 112 728, Mar. 2026, ISSN: 0165-5876.  DOI: 10.1016/j.ijporl.2026.112728. (visited on 05/28/2026).
- 2 K. K. D. Tan, A. M. Condon, C. A. Renteria, *et al.*, “Investigating Cellular Magnetic Bioeffects Using Two-Channel, Two-Photon Autofluorescence Lifetime Microscopy,” *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*, vol. 12, pp. 398–406, 2026, ISSN: 2332-7804.  DOI: 10.1109/TMBMC.2026.3666858. (visited on 05/28/2026).
- 3 J. Shi, **A. Ho**, C. E. Snyder, *et al.*, “Accelerating biopharmaceutical cell line selection with label-free multimodal nonlinear optical microscopy and machine learning,” *en, Communications Biology*, vol. 8, no. 1, pp. 1–15, Feb. 2025, Publisher: Nature Publishing Group, ISSN: 2399-3642.  DOI: 10.1038/s42003-025-07596-w. (visited on 04/27/2025).
- 4 K. K. D. Tan, A. De la Cadena, E. Aksamitiene, **A. Ho**, and S. A. Boppart, “Simultaneous Label-Free Autofluorescence Multi-Harmonic Microscopy,” *Journal of visualized experiments : JoVE*, no. 222, pp. 10.3791/68637, Aug. 2025, ISSN: 1940-087X.  DOI: 10.3791/68637. (visited on 01/05/2026).
- 5 T. T. Roh, A. Alex, P. M. Chandramouleeswaran, *et al.*, “Predicting DNA damage response in non-small cell lung cancer organoids via simultaneous label-free autofluorescence multiharmonic microscopy,” *Redox Biology*, vol. 75, p. 103 280, Sep. 2024, ISSN: 2213-2317.  DOI: 10.1016/j.redox.2024.103280. (visited on 04/27/2025).
- 6 G. L. Monroy, S. T. Fitzgerald, A. Locke, *et al.*, “Multimodal Handheld Probe for Characterizing Otitis Media — Integrating Raman Spectroscopy and Optical Coherence Tomography,” *Frontiers in Photonics*, vol. 3, Jun. 2022, 2024-04-19, ISSN: 2673-6853.  DOI: 10.3389/fphot.2022.929574.
- 7 B. A. Kuzma, I. J. Pence, D. A. Greenfield, **A. Ho**, and C. L. Evans, “Visualizing and quantifying antimicrobial drug distribution in tissue,” *Advanced Drug Delivery Reviews*, vol. 177, p. 113 942, Oct. 2021, 2024-04-19, ISSN: 0169-409X.  DOI: 10.1016/j.addr.2021.113942.
- 8 B. A. Kuzma, I. J. Pence, **A. Ho**, and C. L. Evans, “Visualizing and Quantifying Pharmaceutical Compounds within Skin using Coherent Raman Scattering Imaging,” *JoVE (Journal of Visualized Experiments)*, no. 177, e63264, Nov. 2021, 2024-04-19, ISSN: 1940-087X.  DOI: 10.3791/63264.
- 9 F. Peng, S. Jeong, G. Gonzalez, *et al.*, “Assessment of Glial Fibrillary Acidic Protein Binding to the Surface of Leukocytes with Dark-Field Imaging and Computational Analysis,” *Advanced Functional Materials*, vol. 31, no. 11, p. 2 009 229, 2021, 2024-04-19, ISSN: 1616-3028.  DOI: 10.1002/adfm.202009229.
- 10 F. Peng, S. Jeong, **A. Ho**, and C. L. Evans, “Recent progress in plasmonic nanoparticle-based biomarker detection and cytometry for the study of central nervous system disorders,” *Cytometry Part A*, vol. 99, no. 11, pp. 1067–1078, 2021, 2024-04-19, ISSN: 1552-4930.  DOI: 10.1002/cyto.a.24489.
- 11 S. Jeong, G. González, **A. Ho**, *et al.*, “Plasmonic Nanoparticle-Based Digital Cytometry to Quantify MUC16 Binding on the Surface of Leukocytes in Ovarian Cancer,” *ACS Sensors*, vol. 5, no. 9, pp. 2772–2782, Sep. 2020, 2024-04-19.  DOI: 10.1021/acssensors.0c00567.

Conference Presentations

- 1 M. A. Modi, **A. Ho**, S. T. Fitzgerald, *et al.*, “In vivo clinical evaluation of otitis media with a Raman spectroscopy and optical coherence tomography probe-based system,” in *Imaging, Therapeutics, and Advanced Technology in Head and Neck Surgery and Otolaryngology 2026*, vol. 13825, SPIE, Mar. 2026, p. 1 382 504.  DOI: 10.1117/12.3078915. (visited on 05/28/2026).

- 2 Z. Yang, J. Shi, H. Fan, **A. Ho**, E. Aksamitiene, and S. A. Boppart, "Quantitative phase imaging and nonlinear simultaneous label-free autofluorescence multiharmonic microscopy for structural and functional imaging of cells and tissues," in *Multimodal Biomedical Imaging XXI*, vol. PC13841, SPIE, Mar. 2026, PC1384102.  DOI: 10.1117/12.3079947. (visited on 05/28/2026).
- 3 **A. Ho**, S. Fitzgerald, G. L. Monroy, *et al.*, "Hand-held probe integrating Raman spectroscopy and optical coherence tomography for in vivo characterization of otitis media," in *Imaging, Therapeutics, and Advanced Technology in Head and Neck Surgery and Otolaryngology 2025*, vol. PC13294, SPIE, Mar. 2025, PC1329402.  DOI: 10.1117/12.3043585. (visited on 04/27/2025).
- 4 **A. Ho**, J. Shi, E. Aksamitiene, *et al.*, "Spatial, metabolic, and chemical imaging for enhanced characterization and optimal selection of CHO cell lines via multimodal label-free microscopy," in *Label-free Biomedical Imaging and Sensing (LBIS) 2025*, vol. PC13331, SPIE, Mar. 2025, PC133310A.  DOI: 10.1117/12.3043649. (visited on 04/27/2025).
- 5 G. L. Monroy, Z. Wu, E. J. Chaney, *et al.*, "Cold microplasma exposure as a novel therapeutic treatment for bacterial otitis media: Study results and conclusions in a small animal model," in *Photonic Diagnosis, Monitoring, Prevention, and Treatment of Infections and Inflammatory Diseases 2025*, vol. PC13298, SPIE, Mar. 2025, PC132980H.  DOI: 10.1117/12.3040418. (visited on 04/27/2025).
- 6 G. L. Monroy, Z. Wu, E. J. Chaney, *et al.*, "Cold microplasma exposure as a novel therapeutic treatment: Future implications for the management of bacterial otitis media," in *Imaging, Therapeutics, and Advanced Technology in Head and Neck Surgery and Otolaryngology 2025*, vol. PC13294, SPIE, Mar. 2025, PC1329401.  DOI: 10.1117/12.3040556. (visited on 04/27/2025).
- 7 J. Shi, K. K. D. Tan, **A. Ho**, *et al.*, "DeepFLR: A self-supervised deep learning framework for robust FLIM signal restoration under photon-starved conditions," in *Multiphoton Microscopy in the Biomedical Sciences XXV*, vol. PC13324, SPIE, Mar. 2025, PC1332416.  DOI: 10.1117/12.3043528. (visited on 04/27/2025).
- 8 J. E. Sorrells, **A. Ho**, R. R. Iyer, *et al.*, "Design and implementation of an advanced simultaneous label-free autofluorescence multiharmonic (SLAM) microscopy system for multimodal imaging (Conference Presentation)," in *Multimodal Biomedical Imaging XX*, vol. PC13309, SPIE, Mar. 2025, PC1330901.  DOI: 10.1117/12.3040857. (visited on 04/27/2025).
- 9 K. K. Tan, C. A. Renteria, R. R. Iyer, *et al.*, "Dynamic imaging of magnetic bioeffects in cells using multiphoton autofluorescence intensity and lifetime microscopy," in *Quantum Effects and Measurement Techniques in Biology and Biophotonics II*, vol. PC13340, SPIE, Mar. 2025, PC133400C.  DOI: 10.1117/12.3040978. (visited on 04/27/2025).
- 10 **A. Ho**, Q. Jiang, K. Peterson, *et al.*, "Prognostic value of optical coherence tomography (OCT) with machine learning for assessing clinical outcomes of otitis media," in *Imaging, Therapeutics, and Advanced Technology in Head and Neck Surgery and Otolaryngology 2024*, 2024-04-19, vol. PC12818, SPIE, Mar. 2024.  DOI: 10.1117/12.3003143.
- 11 **A. Ho**, J. Shi, E. Chaney, *et al.*, "Predicting productivity in CHO cells based on co-occurrence of metabolism-related features in simultaneous label-free autofluorescence multiharmonic (SLAM) microscopy," in *Label-Free Biomedical Imaging and Sensing (LBIS) 2024*, 2024-04-19, vol. PC12854, SPIE, Mar. 2024, PC1285407.  DOI: 10.1117/12.3003094.
- 12 J. Shi, **A. Ho**, K. Tan, *et al.*, "Self-supervised deep learning enables robust fluorescence lifetime estimation with limited photons," 2024-04-19, Mar. 2024.
- 13 K. K. Tan, C. A. Renteria, R. R. Iyer, *et al.*, "Probing magnetic field effects on cellular redox state using autofluorescence intensity and lifetime microscopy," in *Quantum Effects and Measurement Techniques in Biology and Biophotonics*, 2024-04-19, vol. PC12863, SPIE, Mar. 2024, PC1286306.  DOI: 10.1117/12.3003226.

- 14 **A. Ho**, Q. Jiang, K. Peterson, *et al.*, “Enhancing diagnostic accuracy of optical coherence tomography (OCT) with machine learning for detection and monitoring of otitis media,” in *22nd International Symposium on Recent Advanced in Otitis Media*, Milwaukee, WI, Jun. 2023.
- 15 **A. Ho**, G. L. Monroy, E. Aksamitiene, D. R. S. Jr, M. Marjanovic, and S. A. Boppart, “Imaging and assessment of human ear wax with optical coherence tomography,” in *Imaging, Therapeutics, and Advanced Technology in Head and Neck Surgery and Otolaryngology 2023*, 2024-04-19, vol. 12354, SPIE, Mar. 2023, p. 1 235 402.  DOI: 10.1117/12.2649235.
- 16 **A. Ho**, J. Shi, E. Chaney, *et al.*, “Characterization of Chinese hamster ovary (CHO) cell lines by simultaneous label-free autofluorescence multi-harmonic (SLAM) microscopy for upstream bioprocessing (Conference Presentation),” in *Imaging, Manipulation, and Analysis of Biomolecules, Cells, and Tissues XXI*, 2024-04-19, vol. PC12383, SPIE, Mar. 2023, PC1238302.  DOI: 10.1117/12.2646378.
- 17 G. L. Monroy, S. T. Fitzgerald, A. Locke, *et al.*, “Integrated Raman spectroscopy and optical coherence tomography system for characterizing otitis media,” in *Imaging, Therapeutics, and Advanced Technology in Head and Neck Surgery and Otolaryngology 2023*, 2024-04-19, vol. 12354, SPIE, Mar. 2023, p. 1 235 404.  DOI: 10.1117/12.2648504.
- 18 G. L. Monroy, J. Kim, **A. Ho**, *et al.*, “Microplasma exposure as a novel therapeutic treatment for otitis media demonstrated in a pre-clinical chinchilla model,” in *Photonic Diagnosis, Monitoring, Prevention, and Treatment of Infections and Inflammatory Diseases 2023*, 2024-04-19, vol. PC12358, SPIE, Mar. 2023, PC1235809.  DOI: 10.1117/12.2648469.
- 19 J. Shi, **A. Ho**, C. E. Snyder, *et al.*, “Machine learning-assisted label-free multimodal optical bioimaging for biopharmaceutical CHO cell line characterization,” in *Multimodal Biomedical Imaging XVIII*, 2024-04-19, vol. PC12371, SPIE, Mar. 2023, PC123710D.  DOI: 10.1117/12.2648499.
- 20 A. Ho, E. H. Rizza, J. Jeskey, *et al.*, “574 A deep learning classifier for the analysis of tiger tail banding in trichothiodystrophy,” 5, 2024-04-19, vol. 141, Elsevier, 2021, S100.

Book Chapters

- 1 A. Juhong, J. Shi, **A. Ho**, G. L. Monroy, and S. A. Boppart, “Methods in molecular biology,” in Springer Nature Link, submitted 2026, ch. Artificial Intelligence in label-free optical imaging applications.
- 2 G. L. Monroy, A. Marckowicz, **A. Ho**, M. Govande, D. R. Spillman, and S. A. Boppart, “Optical coherence tomography: Technology and applications, third edition,” in Springer Reference, submitted 2024, ch. Advanced Optical Coherence Tomography Imaging for Primary Care Medicine.

Research Summary

2021 – Present

Graduate Research Assistant: Boppart Lab

Beckman Institute for Advanced Science and Technology,
University of Illinois Urbana-Champaign (Urbana, IL, United States)

- Development of a combined Raman Spectroscopy and Optical Coherence Tomography (OCT) probe and portable cart for diagnosis of otitis media.
- Reader study for clinical comparison between B-mode and M-mode OCT for diagnosis of otitis media
- Biomarker discovery of cholesteatoma using OCT
- Development of catheter based imaging probes for OCT imaging of ear canal tissue and wax for Proctor and Gamble
- Collaboration with GlaxoSmithKline (GSK) on Non-invasive and Non-Destructive Characterization of genetically modified functional Chinese Hamster Ovary cells.
- Collaboration with GSK to investigate methods for quantifying cilia beating frequency through imaging
- Collaboration with GSK to investigate Antisense Oligonucleotide (ASO) drug delivery in hepatocytes with coherent Raman imaging.
- Development of an automated, multimodal, multiphoton laser scanning microscope for longitudinal imaging. Wrote custom python-based acquisition software for synchronization of hardware components during acquisition.

2018 – 2021

Research Technician II: Evans Lab

Wellman Center for Photomedicine, Massachusetts General Hospital,
Harvard Medical School (Boston, MA, United States)

- Design and development of a non-linear optical imaging system for studying the pharmacodynamics and pharmacokinetics of topically applied drugs.
- Design and development of a nanoparticle based method for the ultrasensitive detection of leukocyte-ligand interactions in ovarian cancer and traumatic brain injury.
- Design and development of a device to enhance platelet shelf life using photobiomodulation.
- Development of a deep learning image classifier to assist in diagnosis of Xeroderma Pigmentosum and Trichothiodystrophy.

2019 – 2021

Visiting Research Collaborator: Sorger Lab

The Center for Cancer Systems Pharmacology (CCSP),
Harvard Medical School (Boston, MA, United States)

- Development of a Coherent Raman Microscope for Multiplexed Immunohistochemical Analysis.

Research Summary (continued)

2015 – 2018

📌 **Undergraduate Researcher: Georgakoudi Lab**

Tufts University, School of Engineering,
Department of Biomedical Engineering (Medford, MA, United States)

- Development of a differentially polarized light laparoscope for detection of peritoneal cancer.

Teaching

Teaching

2023 – 2026

📌 **CLIMB Summer School Teaching Assistant**

- Teaching assistant for two, 2 week intensive summer course on biophotonics at the University of Illinois Urbana-Champaign for graduate students and postdoctoral scholars from across the world.
- Development of course curriculum and workshops for pulse characterization of ultrafast lasers, optical tweezers, and GPU acceleration with Python
- Taught introductory machine learning and deep learning lab to students with Python
- Taught lab for translation of imaging technologies to the clinic, integrating hardware (optics), software (Labview, Python, CUDA), and human factors engineering into a portable Optical Coherence Tomography (OCT) cart for clinical imaging.

Mentoring

2020

📌 **Viktoria Catalan.** U-Net denoising for Coherent Raman Microscopy as part of HST-Wellman Summer Institute for Biomedical Optics

2016

📌 **Various High School Students.** Computer Aided Design for Laparoscope Design as part of Tufts TUBERS

Service

Professional Membership

2022 – 2026

📌 **IEEE Photonics UIUC Student Chapter:** Treasurer

📌 **Optica UIUC Student Chapter:** Treasurer

📌 **SPIE UIUC Student Chapter:** Treasurer

Academic Service

2023 – present

📌 **Peer Reviewer for Academic Journals**

- Frontiers in Neuroscience
- PLOS ONE

Service (continued)

Outreach Activities

2026 **TiME T32 Annual Symposium Support**

- Assisted in organization and coordination of the TiME NIH T32 annual symposium at the University of Illinois Urbana-Champaign.
- Documented symposium activities through event photography and media support to promote trainee engagement and scientific communication.

2024 – 2026 **CLIMB Summer School Media Support**

- Provided photography and media documentation for the NIH P41 CLIMB Summer School, including trainee activities, lectures, laboratory demonstrations, and outreach events.
- Produced visual media used in program newsletters, social media, outreach materials, and institutional communications.

2024 – 2025 **Tuscola Middle School Outreach**

- Gave lectures and led interactive demonstrations at East Prairie Middle School in Tuscola, IL on ear imaging including topics such as hearing, ear infections, and diagnosis of ear infections with OCT.

2024 **CLIMB Booth Representative**

- Represented the NIH P41 center, Center for Label-free Imaging and Multiscale Biophotonics (CLIMB), at SPIE Photonics West.
- Explained resources provided by the center to conference attendees such as service projects, summer schools, invited speaker programs, etc.

2023 – 2026 **Beckman Open House: Lab Outreach Coordinator**

- Organized two open house events for the Boppart lab, coordinating graduate students and post-doctoral researchers in presenting interactive science demonstrations and educational presentations for label-free imaging to students of all ages and other event attendees.
- Engaged with visitors of all ages to explain scientific concepts and theories in an accessible manner.
- Performed demonstrations of retinal and ear imaging to illustrate principles of optical coherence tomography

Service (continued)

2022 – 2026

📌 iOptics Executive Board Treasurer

- Coordinated and invited optics related speakers (22 internal and 4 external) to the University of Illinois Urbana-Champaign.
- Hosted external speakers from other universities in the United States, Canada, France, and Taiwan
- Managed funds and generated financial and tax reports for professional societies (IEEE Photonics, Optica, SPIE)
- Organized lunches for biweekly seminars

Skills

Languages	📌	English (native), Cantonese (native, speaking), Spanish (intermediate)
Coding	📌	Python, Matlab, R, C++, CUDA, $\text{\LaTeX}$, Deep Learning (Tensorflow, Pytorch), Git
Biology	📌	Cell Culture, Immunoassays, Animal Handling (mice and chinchilla), Blood Processing, Skin Processing
Engineering	📌	PCB design (Eagle), CAD (Fusion360, Solidworks), 3D-printing (FDM, SLA), Soldering (micro, reflow), LabVIEW (DAQmx) for timing and synchronization
Optics	📌	Alignment, Pulse Characterization (Autocorrelation, FROG, MIIPS), Pulse Shaping (SLM), Fiber Optics
Imaging	📌	Brightfield Microscopy, Dark Field Microscopy, CARS Microscopy, SRS Microscopy, Multiphoton Microscopy, Optical Coherence Tomography, Fluorescence Microscopy

Miscellaneous Experience

Awards and Achievements

2025	📌	Tissue Microenvironment (TiME) Training Program Fellowship
2024	📌	McGinnis Medical Innovation Fellowship
2023	📌	Travel Award , 22nd Symposium on Recent Advances in Otitis Media (ISOM 2023)
2018	📌	Magna Cum Laude , Tufts University
	📌	Dean's List 8/8 Semesters , Tufts University
2017	📌	Kullman Scholar , Tufts University

Certifications

2016 – 2019	📌	EMT B. Awarded by National Registry of Emergency Medical Technicians
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References

Available on Request